THE CONSTANT SHEAF ON AN IRREDUCIBLE TOPOLOGICAL SPACE HAS NO HIGHER COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and is four sentences long in the source; nothing is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-irreducible-constant-cohomology-zero is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Definition of the constant sheaf. Sheaves, Definition sheaves-definition-constant-sheaf, invoked at proof lines 3469-3470. Supplies the description of the presheaf that has to be shown to be flasque.
- Flasque modules are acyclic for the global section functors. Same chapter, Lemma lemma-flasque-acyclic, invoked at proof lines 3475-3476. Supplies the vanishing of higher cohomology.
- Nonempty open subsets of an irreducible space are irreducible and connected. Standard topology, used at proof lines 3471-3473. Supplies that all the restriction maps of the constant sheaf on nonempty opens are isomorphisms, hence flasqueness.

Source reference. Original lemma 09SY: sources/stacks-cohomology/pinned-original-source.tex, source lines 1913–1913. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 006W: sources/stacks-chow/tags.txt, source lines 260–260. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be an irreducible topological space. Then $H^p(X, \underline{A}) = 0$ for all $p > 0$ and any abelian group A.*

Proof. Recall that $\underline{A}$ is the constant sheaf as defined in Sheaves, Definition 006W. Since X is irreducible, any nonempty open U is irreducible and a fortiori connected. Hence for $U \subset X$ nonempty open we have $\underline{A}(U) = A$. We have $\underline{A}(\emptyset) = 0$. Thus $\underline{A}$ is a flasque abelian sheaf on X. The vanishing follows from Lemma 09SY. □

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